Spectral Mobility and the Lorentz Factor from Projective Capacity Saturation

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Abstract

The O-series saturation parameter η_n^{O7} is defined in [3] as the ratio of cumulative occupied projective capacity to total residual frontier novelty. The analytic factorization proved below explains why this total ratio satisfies a linear filling law $B_n + \Sigma_n^{\mathrm{tot}} = 1$ and does not by itself yield the Lorentz factor: $\omega_q(a, b, c) = e^{2\pi i c/q}\omega_q(a, b, 0)$, so the central coordinate c contributes only a global phase and is invisible to the fingerprint span. We identify the conditional mechanism by which a temporally projected residual $\Sigma_n^\tau := \mathcal{P}_\tau^{\mathrm{Q5b}}\,\Sigma_n^{\mathrm{tot}}$ can produce the relativistic mobility ratio: under the Born–Infeld admissibility budget established in [2], a quadratic normalization of B_n and Σ_n^τ implies

$$\eta_n^\tau := \frac{B_n}{\Sigma_n^\tau} \longrightarrow \beta\gamma,$$

where $\beta\gamma$ is the relativistic mobility and γ is the Lorentz factor. Consequently, $\Phi(\eta_n^\tau) = 1/\sqrt{1 + (\eta_n^\tau)^2}$ converges to $1/\gamma$. The temporal projection operator $\mathcal{P}_\tau^{\mathrm{Q5b}}$ is induced by the Carnot–Carathéodory geometry and co-metric closure of [1, 4]. Constructing this operator and verifying $\mathcal{P}_\tau^{\mathrm{Q5b}}\Sigma_n^{\mathrm{tot}} \to \sqrt{1 - B_n^2}$ is the remaining technical task.

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1 O7 occupation and residual novelty

The O7 saturation parameter is defined in [3] by

$$\eta_n^{\mathrm{O7}}(x) = \frac{B_n(x)}{\Sigma_n(x)}, \qquad B_n(x) := |\psi_n(x)|^2 = \frac{|S_n \cap C_x|}{|C_x|}.$$

Here C_x is the effective resolution cell associated with the observable state x, and S_n is the BFS-reachable set up to cascade depth n.

The observable $B_n(x)$ is cumulative. It measures the fraction of the cell C_x already reached by the BFS cascade by depth n. In particular,

$$0 \le B_n(x) \le 1, \qquad B_{n+1}(x) \ge B_n(x).$$

It is therefore an occupation observable, not an infinitesimal mobility.

By contrast, $\Sigma_n(x)$ measures the average effective novelty contributed by the BFS frontier at depth n: the mean fraction of genuinely new projective direction brought by admissible paths reaching ∂S_n within C_x. It is a residual frontier observable, instantaneous and non-increasing in n.

The ratio η_n^{O7} is therefore not a fitting parameter. It is an occupied-over-residual capacity ratio:

$$\eta_n^{\mathrm{O7}} = \frac{\text{cumulative occupied capacity}}{\text{residual frontier novelty}}.$$

The question addressed in this note is whether this O7 ratio converges, in the continuum geometric limit, to the relativistic mobility ratio $dI_{\mathrm{space}}/dI_\tau$.

2 Algebraic Born–Infeld target

The cascade depth n is not a time coordinate. It is the ordinal depth of an admissible projective history, counting the number of distinguishable projective transitions undergone along that history [2].

Assume a bounded Born–Infeld mobility budget at each projective depth increment dn:

$$dI_\tau^2 + dI_{\mathrm{space}}^2 = c_{\mathrm{BI}}^2\, dn^2.$$

Define the normalized spectral mobility

$$\beta := \frac{dI_{\mathrm{space}}}{c_{\mathrm{BI}}\, dn}.$$

Then the residual temporal component is

$$dI_\tau = c_{\mathrm{BI}} \sqrt{1-\beta^2}\, dn,$$

and therefore

$$\frac{dI_{\text{space}}}{dI_\tau} = \frac{\beta}{\sqrt{1-\beta^2}} = \beta\gamma, \qquad \gamma := \frac{1}{\sqrt{1-\beta^2}}.$$

If a spectral saturation parameter η satisfies $\eta = dI_{\text{space}}/dI_\tau$, then

$$\Phi(\eta) = \frac{1}{\sqrt{1+\eta^2}} = \frac{1}{\gamma}.$$

The inverse Lorentz factor is the residual-capacity profile of a quadratic Born–Infeld budget. This step is purely algebraic. The non-trivial question is whether the O7 total residual Σ_n contains a temporal component Σ_n^τ that converges to the normalized temporal budget component $\sqrt{1-\beta^2}$.

3 Why the reduced filling model is insufficient

The reduced filling model of [3] gives, before geometric correction,

$$B_n \simeq \frac{n}{D}, \qquad \Sigma_n \simeq 1 - \frac{n}{D},$$

equivalently $B_n + \Sigma_n \simeq 1$. This is a linear residual law, describing monotone filling of a finite resolution cell.

Numerical confirmation. Direct computation on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for $q \in \{7, 11, 13\}$, using cumulative (a, b) spatial coverage for B_n and the total residual Weil-representation span for Σ_n^{tot}, yields mean deviations:

$$|B_n + \Sigma_n^{\text{tot}} - 1| \approx 0.02\text{–}0.04, \qquad |B_n^2 + (\Sigma_n^{\text{tot}})^2 - 1| \approx 0.17\text{–}0.26.$$

The data confirm the linear law for the total residual. The quadratic law fails by a factor of approximately $1/\sqrt{1-B_n^2}$ relative to the linear law, indicating that Σ_n^{tot} is too small by this factor to be the Lorentzian residual.

Interpretation. The O7 total residual Σ_n^{tot} measures total Weil-space coverage and satisfies $B_n + \Sigma_n^{\text{tot}} \simeq 1$. The Lorentzian residual factor $\sqrt{1-\beta^2}$ is not obtained from Σ_n^{tot} directly. It requires projecting Σ_n^{tot} onto the *temporal component* selected by the Q5b principal-symbol geometry:

$$\Sigma_n^\tau := \mathcal{P}_\tau^{\text{Q5b}}\, \Sigma_n^{\text{tot}},$$

where $\mathcal{P}_\tau^{\text{Q5b}}$ is the temporal residual projection induced by the completed effective co-metric. The corrected quadratic target is

$$B_n^2 + (\Sigma_n^\tau)^2 \longrightarrow 1.$$

4 From linear filling to quadratic normalization via Q5b

4.1 Q5b input

The reduced filling model operates at the combinatorial level of a finite resolution cell. The Q5b continuum limit supplies the missing geometric structure [1].

In the pre-saturation large-q regime, BFS shells on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ converge to Carnot–Carathéodory (CC) balls on $\mathrm{Heis}_3(\mathbb{R})$. The linear residual law of the reduced model is a coarse filling approximation valid at finite D and finite q; it does not encode the sub-Riemannian anisotropy of the CC geometry.

The effective co-metric extracted from the completed principal-symbol geometry of the admissibility operator is Lorentzian in the homogeneous regime. Assuming the Q11 co-metric closure [4], the normalized co-metric is

$$g^{\mu\nu} = 2\,\eta^{\mu\nu}, \qquad \eta = \mathrm{diag}(-1, +1, +1, +1),$$

with no free parameter. This Lorentzian structure provides the quadratic normalization in which the continuum admissible components are to be read, replacing the linear complement of the reduced filling model. The remaining task is to identify the O7 observables with these normalized components.

The chain is therefore:

CC convergence (Q5b)	$\Rightarrow$	continuum spatial and temporal components exist,
principal symbol / Q11 closure	$\Rightarrow$	temporal projection operator $\mathcal{P}_\tau^{\mathrm{Q5b}}$ defined,
Lemma 1	$\Rightarrow$	$B_n^2 + (\mathcal{P}_\tau^{\mathrm{Q5b}}\Sigma_n^{\mathrm{tot}})^2 \to 1$.

The geometric structure is not postulated; it comes from Q5b and Q11. The remaining task is to construct $\mathcal{P}_\tau^{\mathrm{Q5b}}$ explicitly and verify that the normalized temporal component extracted from Σ_n^{tot} converges to $\sqrt{1 - B_n^2}$.

4.2 Main convergence lemma

[Q5b temporal residual projection] Let Σ_n^{tot} denote the O7 total residual frontier novelty, viewed in the large-q limit as a residual section of the continuum admissible capacity bundle over the Q5b Carnot limit. The Q5b temporal residual projection is the component map

$$\mathcal{P}_\tau^{\mathrm{Q5b}} : \Sigma_n^{\mathrm{tot}} \mapsto \Sigma_n^{\tau}$$

induced by the completed effective co-metric $g^{\mu\nu} = 2\,\eta^{\mu\nu}$ of [1, 4]. It is defined by projecting the total residual admissible capacity onto the component orthogonal, in the completed Lorentzian co-metric, to the occupied spatial component B_n.

The direction identified as *temporal* here is temporal only after the chain

$$\text{Carnot structure} \;\to\; \text{symbol completion} \;\to\; \text{Q11 co-metric closure} \;\to\; g^{\mu\nu} = 2\,\eta^{\mu\nu}.$$

$\mathcal{P}_\tau^{\mathrm{Q5b}}$ is therefore *not* the naive projector onto the central direction of the sub-Laplacian Δ_H; the central direction Z appears at sub-principal / Carnot degree 2 via $[X, Y] = Z$, and becomes a metric temporal direction only after co-metric completion.

Lemma 1 (Quadratic residual component)**.** *In the homogeneous pre-saturation regime, let $B_n(x) := |\psi_n(x)|^2$ be the O7 cumulative occupation observable and $\Sigma_n^{\mathrm{tot}}(x)$ the O7 total residual frontier novelty [3]. Direct computation confirms the linear filling law:*

$$B_n + \Sigma_n^{\mathrm{tot}} \simeq 1.$$

Let $\Sigma_n^\tau := \mathcal{P}_\tau^{\mathrm{Q5b}}\,\Sigma_n^{\mathrm{tot}}$ be the temporal component of Definition 1. The decomposition

$$\Sigma_n^{\mathrm{tot}} = \Sigma_n^{\mathrm{spat}} \oplus \Sigma_n^\tau$$

induced by the completed co-metric separates the total residual into spatial and temporal components. If the Q5b observable identification sends B_n to the normalized spatial component of the completed Born–Infeld budget, then

$$B_n^2 + (\Sigma_n^\tau)^2 \longrightarrow 1,$$

equivalently $\Sigma_n^\tau \to \sqrt{1-\beta^2}$, where

$$\beta := \frac{dI_{\mathrm{space}}}{c_{\mathrm{BI}}\,dn}$$

is the normalized spectral mobility, represented after physical normalization $c_{\mathrm{BI}} \mapsto c$ as v_{eff}/c.

The remaining work follows the decomposition chain:

$$\begin{aligned}
&\Sigma_n^{\mathrm{tot}} \quad \text{(O7 total residual)}\\
&\quad\downarrow \quad \text{Q5b decomposition of admissible components}\\
&\Sigma_n^{\mathrm{tot}} = \Sigma_n^{\mathrm{spat}} \oplus \Sigma_n^\tau\\
&\quad\downarrow \quad \text{BI/Q11 normalization}\\
&B_n^2 + (\Sigma_n^\tau)^2 \to 1.
\end{aligned}$$

This is an identification between the O7 total observable and the Q5b/Q11 geometric decomposition, not a new geometric conjecture.

5 Weil factorization and the invisibility of the central direction

5.1 The Schrödinger representation and its factorization

The Schrödinger (Weil) representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ acts on $\ell^2(\mathbb{Z}/q\mathbb{Z}) \cong \mathbb{C}^q$ by

$$[\omega_q(a,b,c)\,\psi](x) = e^{2\pi i(c+ax)/q}\,\psi(x+b \bmod q).$$

[Central factorization] For all $(a,b,c) \in \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$,

$$\omega_q(a,b,c) = e^{2\pi ic/q} \cdot \omega_q(a,b,0).$$

Proof. Direct from the formula: $[\omega_q(a,b,c)\psi](x) = e^{2\pi ic/q} \cdot e^{2\pi iax/q}\psi(x+b) = e^{2\pi ic/q} \cdot [\omega_q(a,b,0)\psi](x)$. □

5.2 Consequence: central direction is invisible to span

Proposition 1 has an immediate corollary for the fingerprint span.

Corollary 1 (Linear filling law)**.** *The span of* $\{\omega_q(g) : g \in \bigcup_{k\le n} S_k\}$ *in* $\mathrm{End}(\mathbb{C}^q)$ *depends only on the projected* (a,b) *coordinates:*

$$\mathrm{span}\{\omega_q(a,b,c)\} = \mathrm{span}\{\omega_q(a,b,0)\}.$$

The span dimension at depth n *equals the number of distinct* (a,b) *pairs in* $\bigcup_{k\le n} S_k$*. On* $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ *with generators* $\{(\pm1,0,0),(0,\pm1,0)\}$*, the BFS explores the* L^1 *ball on* $(\mathbb{Z}/q\mathbb{Z})^2$*:*

$$B_n = \frac{|\{(a,b) : \min(a,q-a) + \min(b,q-b) \le n\}|}{q^2} = \frac{2n^2+2n+1}{q^2} \quad (n < q/2),$$

and the total residual novelty satisfies:

$$\Sigma_n^{\mathrm{tot}} = 1 - B_n.$$

The linear law $B_n + \Sigma_n^{\mathrm{tot}} = 1$ is therefore *exact* in the Schrödinger representation, not an approximation. The central coordinate c is invisible to the fingerprint span: it contributes only a global phase $e^{2\pi ic/q}$, which does not change the span direction.

5.3 Why this does not obstruct the Lorentz identification

The factorization shows that the temporal direction required by the Lorentz identification is *not* directly visible in the Schrödinger representation. This is expected: in Carnot–Carathéodory geometry, the central direction Z appears at degree 2 via the bracket $[X, Y] = Z$, not at degree 1 alongside the horizontal generators. It becomes a metric temporal direction only after the symbol completion chain

$$\text{CC structure} \;\to\; \text{sub-principal symbol correction} \;\to\; \text{Q11 co-metric closure} \;\to\; g^{\mu\nu} = 2\,\eta^{\mu\nu}.$$

Consequently, the projection $\mathcal{P}_\tau^{\mathrm{Q5b}}$ of Definition 1 does *not* act on the Schrödinger representation space $\mathbb{C}^q$ directly. It acts on the *capacity functional* Σ_n^{tot} viewed as a section of the admissible capacity bundle over the Q5b Carnot limit, projecting onto the component induced by the completed co-metric.

The factorization result therefore clarifies the structure of the remaining proof: one must go beyond the Schrödinger representation and work at the level of the Q5b principal-symbol geometry to extract the temporal residual Σ_n^τ.

6 Numerical tests for candidates of Σ_n^τ

6.1 Setup and candidates

The Weil factorization of Section 5 identifies the central coordinate c as invisible to the fingerprint span. The projection $\mathcal{P}_\tau^{\mathrm{Q5b}}$ must therefore extract the temporal residual from outside the Schrödinger representation. To probe what kind of observable could fill this role, we test three natural candidates on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for $q \in \{11, 13, 17\}$.

With $B_n = (2n^2 + 2n + 1)/q^2$ (proved in Section 5), the candidates are:

- Σ_n^A : residual c-value coverage, $1 - |\{c : \exists (a, b),\ (a, b, c) \in \bigcup_{k \le n} S_k\}|/q$. This is the fraction of central values not yet reached by the BFS.
- $\Sigma_n^B = \sqrt{1 - B_n}$: the square-root complement of spatial coverage. By construction it satisfies $B_n + (\Sigma_n^B)^2 = 1$, so it serves as a reference for the "half-linear" law.
- $\Sigma_n^C = \sqrt{1 - \beta_n^2}$ with $\beta_n = n/n_{\max}$: the velocity-proxy complement, where β_n is the normalized BFS depth.

For each candidate Σ_n^X we measure the quadratic sum $B_n^2 + (\Sigma_n^X)^2$ and compare to 1.

6.2 Results

Table 1 reports the mean deviation $\langle |B_n^2 + (\Sigma_n^X)^2 - 1| \rangle$ over mid-range depths.

q	Σ_n^A (c-coverage)	Σ_n^B ($\sqrt{1-B}$)	Σ_n^C (depth proxy)
11	0.450	0.142	0.075
13	0.460	0.137	0.076
17	0.474	0.132	0.078

Table 1: Mean $|B_n^2 + (\Sigma_n^X)^2 - 1|$ over mid-range BFS depths. All candidates fail the quadratic law.

Σ_n^A (c-coverage). Deviation $\approx$ 45–47%: largest failure. The central coordinate c saturates at depth $n \approx q/2$ (all q values of c are covered by mid-cascade), while B_n is still near 0.5. The two observables run on incompatible scales.

$\Sigma_n^B = \sqrt{1 - B_n}$. Satisfies $B_n + (\Sigma_n^B)^2 = 1$ exactly by construction, but the *quadratic* sum deviates by $\approx 13\%$. This is because $B_n^2 + (1 - B_n) = 1 - B_n(1 - B_n) \neq 1$ unless $B_n \in \{0, 1\}$. It confirms that $B_n + \Sigma^2 = 1$ and $B_n^2 + \Sigma^2 = 1$ are genuinely different conditions.

$\Sigma_n^C = \sqrt{1 - \beta_n^2}$ with $\beta_n = n/n_{\max}$. Closest to the quadratic law ($\approx$ 7–8% deviation), with $B_n^2 + (\Sigma_n^C)^2 \approx 1.004$ near $\beta_n = 0.5$. However, the sum rises to ≈ 1.18 near $\beta_n = 0.8$ before returning to 1 at $n = n_{\max}$. The depth proxy is not the correct velocity coordinate.

6.3 Interpretation

No natural BFS observable satisfies the quadratic law $B_n^2 + (\Sigma_n^\tau)^2 \to 1$ with $B_n = (2n^2 + 2n + 1)/q^2$. The failure modes are structurally distinct: Σ_n^A saturates too fast (scaling mismatch); Σ_n^B satisfies a linear-square law, not a fully quadratic one; Σ_n^C is close but biased in the high-β regime.

This confirms that $\mathcal{P}_\tau^{\mathrm{Q5b}}$ cannot be replaced by any simple function of the BFS cascade observables. The temporal projection must be extracted from the capacity functional at the level of the Q5b completed co-metric, as stated in Definition 1. Constructing this projection explicitly from Q5b §3–4 is the remaining open task.

7 Consequence for the Lorentz factor

Theorem 1 (Lorentz factor from O7 residual capacity, conditional on Lemma 1)**.** *Assume the Born–Infeld mobility budget of Section 2 and the convergence of Lemma 1. Define the temporally projected saturation parameter*

$$\eta_n^\tau := \frac{B_n}{\Sigma_n^\tau} = \frac{B_n}{\mathcal{P}_\tau^{\mathrm{Q5b}}\,\Sigma_n^{\mathrm{tot}}}.$$

Then

$$\eta_n^\tau \longrightarrow \frac{\beta}{\sqrt{1-\beta^2}} = \beta\gamma,$$

and therefore the O-series profile with temporal projection satisfies

$$\Phi(\eta_n^\tau) = \frac{1}{\sqrt{1+(\eta_n^\tau)^2}} \longrightarrow \frac{1}{\gamma}.$$

The special-relativistic residual factor is recovered from the temporally projected O7 ratio B_n/Σ_n^τ. The O7 total ratio $\eta_n^{\mathrm{O7}} = B_n/\Sigma_n^{\mathrm{tot}}$ satisfies the linear law and does not directly yield the Lorentz factor; the Q5b temporal projection is the essential intermediate step.

Remark 1 (Projected saturation pressure at $v \to c$)**.** *For the temporally projected analogue $P_{n,\tau}^{\mathrm{sat}} := \gamma\,\eta_n^\tau$, one obtains $P_{n,\tau}^{\mathrm{sat}} \to \beta\gamma^2$ under $\eta_n^\tau \to \beta\gamma$, which diverges as $v \to c$. This is consistent with the light cone reading $dI_\tau = 0 \Leftrightarrow v = c$: the budget is fully consumed by spatial mobility and no temporal residual remains. Note that the O7 total pressure $P_n^{\mathrm{sat}} = \gamma\,\eta_n^{\mathrm{O7}}$ satisfies the linear law and does not exhibit this divergence directly.*

Remark 2 (Scope)**.** *Theorem 1 directly yields time dilation ($d\tau/dn = 1/\gamma$) and the light cone ($dI_\tau = 0 \Leftrightarrow v = c$). Length contraction requires the additional reconstruction of effective Lorentz transformations between inertial observers. The extension to general relativity is expected to proceed through the non-uniform regime: when localised configurations modify the local spectral budget, the position-dependent metric $g_{\mu\nu}(x)$ encodes gravitational time dilation as a comparison of residual temporal capacities along distinct projected histories.*

References

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